First Steps in Trigonometry

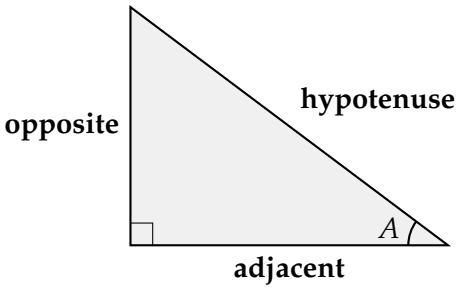
Sine, Cosine and Tangent — the secret ratios inside every right triangle

Answer Key

The big idea

Trigonometry ("triangle measuring") lets you find a length you *can't* measure — the height of a tree, a building, a kite — from one length and one angle you *can* measure.
It works because in a right triangle, **the angle decides the shape**, and the shape decides the **ratios** of the sides.

1 Name the sides — always from the angle you are looking at

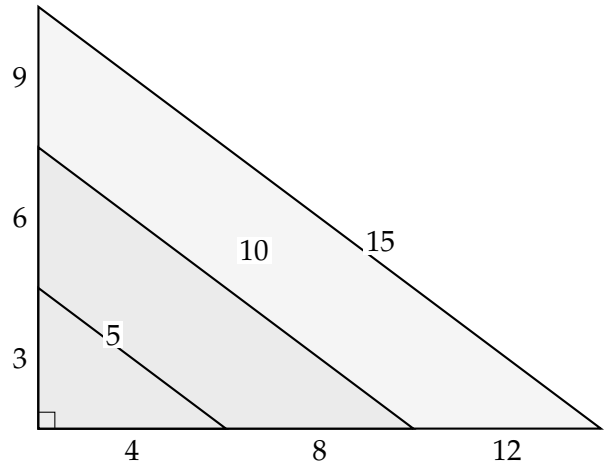


Stand at the angle A (the red arc) and look into the triangle:

- **Hypotenuse** — the longest side, **across from the right angle** ⌞. It never changes.
- **Opposite** — the side **across from** angle A. It doesn't touch A.
- **Adjacent** — the side **next to** angle A that is *not* the hypotenuse. ("Adjacent" means "next to.")

Watch out: opposite and adjacent **swap** if you stand at the other sharp angle. Always ask first: "Which angle am I looking from?"

2 Why ratios? A simple case with three triangles



These are **scale drawings** of one triangle (scale factors 1, 2, 3), so they all share the same angles. Look at $\frac{\text{opposite}}{\text{hypotenuse}}$ for the bottom-right angle:

$$\frac{3}{5} = 0.6 \qquad \frac{6}{10} = 0.6 \qquad \frac{9}{15} = 0.6$$

Same angle $\Rightarrow$ **same ratio**, no matter how big the triangle is. These ratios get names: sine, cosine and tangent.

3 The three definitions: SOH CAH TOA

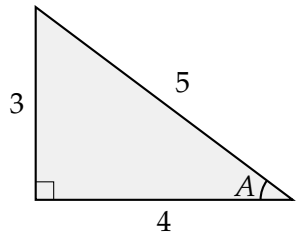
For an acute angle A in a right triangle

$$\sin A = \frac{\textbf{O}\text{pposite}}{\textbf{H}\text{ypotenuse}} \qquad \cos A = \frac{\textbf{A}\text{djacent}}{\textbf{H}\text{ypotenuse}} \qquad \tan A = \frac{\textbf{O}\text{pposite}}{\textbf{A}\text{djacent}}$$

SOH CAH TOA

Memory sentence: *Some Old Horses Can Always Hear Their Owners Approach.*

Example 1 · read the ratios



From angle A: opposite $= 3$, adjacent $= 4$, hypotenuse $= 5$.

$$\sin A = \frac{3}{5} = 0.6 \qquad \cos A = \frac{4}{5} = 0.8 \qquad \tan A = \frac{3}{4} = 0.75$$

Simple check: sine and cosine are always **less than 1**, because the hypotenuse is the longest side.

4 Use a table to find a missing side

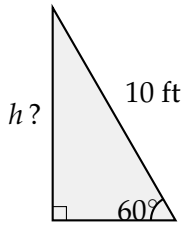
Angle	30°	37°	45°	53°	60°
sin	0.50	0.60	0.71	0.80	0.87
cos	0.87	0.80	0.71	0.60	0.50
tan	0.58	0.75	1.00	1.33	1.73

Values are rounded. (A scientific calculator gives them too — make sure it is in **degree** mode.)

Four steps

1. Mark the angle you know. **2.** Label the sides: hyp, opp, adj. **3.** Pick the ratio that uses the side you *know* and the side you *want*. **4.** Write the equation and solve it like a proportion.

Example 2 · how high does the ladder reach?



A 10-foot ladder leans on a wall at a 60° angle with the ground.
Know: hypotenuse $= 10$. Want: the height, which is **opposite** the 60° angle.
Opposite and hypotenuse $\Rightarrow$ **SOH**, sine:

$$\sin 60^\circ = \frac{h}{10} \;\Rightarrow\; 0.87 = \frac{h}{10} \;\Rightarrow\; h = 10 \times 0.87 = 8.7 \text{ ft.}$$

Example 3 · the unknown is on the bottom

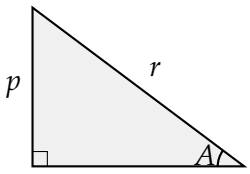
The opposite side of a 30° angle is 6. How long is the hypotenuse? $\sin 30^\circ = \frac{6}{h} \Rightarrow 0.5 = \frac{6}{h} \Rightarrow h = \frac{6}{0.5} = 12$. **Check:** $\frac{6}{12} = 0.5$ ✓

5 Practice

A. Name the sides EASY

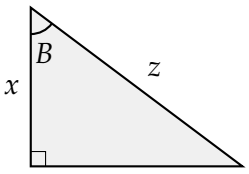
For the marked angle, write which side is the hypotenuse, the opposite and the adjacent.

1.



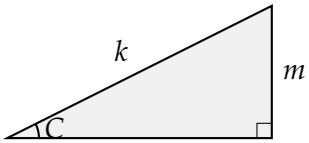
hyp r, opp p, adj q

2.



hyp z, opp y, adj x

3.

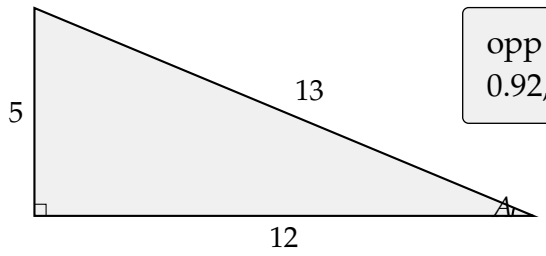


hyp k, opp m, adj n

B. Write the ratios EASY MEDIUM

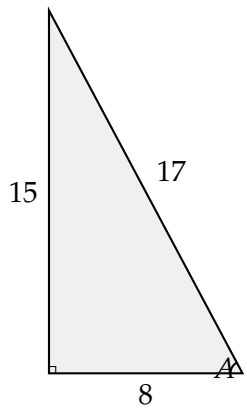
Write $\sin A$, $\cos A$ *and* $\tan A$ *as fractions in lowest terms, then as decimals (round to the hundredths).*

4.



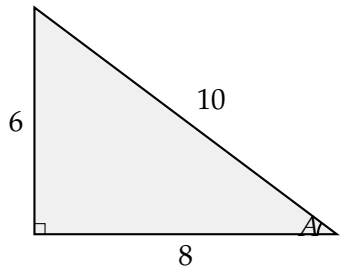
opp 5, adj 12, hyp 13. $\sin A = \frac{5}{13} \approx 0.38$, $\cos A = \frac{12}{13} \approx 0.92$, $\tan A = \frac{5}{12} \approx 0.42$.

5.



opp 15, adj 8, hyp 17. $\sin A = \frac{15}{17} \approx 0.88$, $\cos A = \frac{8}{17} \approx 0.47$, $\tan A = \frac{15}{8} = 1.875 \approx 1.88$.
(A tangent *can* be bigger than 1 — when the opposite side is longer than the adjacent.)

6.



$\sin A = \frac{6}{10} = \frac{3}{5} = 0.6$, $\cos A = \frac{8}{10} = \frac{4}{5} = 0.8$, $\tan A = \frac{6}{8} = \frac{3}{4} = 0.75$.
Same as the 3-4-5 triangle — it is a scale drawing (factor 2), so it has the same angles.

7. MEDIUM In the triangle of problem 4, now stand at the **other** sharp angle B. Write $\sin B$, $\cos B$ and $\tan B$. What do you notice?

From B: opp 12, adj 5, hyp 13. $\sin B = \frac{12}{13}$, $\cos B = \frac{5}{13}$, $\tan B = \frac{12}{5}$.
$\sin B = \cos A$ and $\cos B = \sin A$ — opposite and adjacent swapped. $\tan B$ is the reciprocal of $\tan A$.

C. Same angle, same ratio MEDIUM

8. Triangle P has legs 5 and 12. Triangle Q has legs 10 and 24. Find tan of the angle across from the short leg in each. Are the two angles equal? Explain.

P: $\tan = \frac{5}{12}$. Q: $\tan = \frac{10}{24} = \frac{5}{12}$. Same ratio, so the angles are equal. Q is a scale drawing of P with scale factor 2 — scaling changes lengths, not angles.

D. Which ratio? MEDIUM

Write sin, cos or tan. (Which two sides does the problem use?)

9. Know the hypotenuse, want the opposite.

sin (SOH)

10. Know the adjacent, want the opposite.

tan (TOA)

11. Know the hypotenuse, want the adjacent.

cos (CAH)

12. Know the opposite, want the hypotenuse.

sin (SOH)

E. Find the missing side MEDIUM HARD

Use the table in Part 4. Draw a picture, label hyp / opp / adj, then solve.

13. MEDIUM The hypotenuse is 20 and one angle is 30°. Find the side opposite the 30° angle.

$\sin 30° = \frac{x}{20},\ 0.5 = \frac{x}{20},\ x = 20 \times 0.5 = 10.$

14. MEDIUM A ramp is 10 m long and rises at an angle of 30°. How high does it go?

The ramp is the hypotenuse; the height is opposite: $\sin 30° = \frac{h}{10},\ h = 10 \times 0.5 = 5$ m.

15. MEDIUM One leg is 8 and the angle next to it is 45°. Find the other leg (opposite the 45° angle).

Adjacent 8, want opposite: $\tan 45° = \frac{x}{8},\ 1 = \frac{x}{8},\ x = 8.$ (A 45° right triangle has two equal legs.)

16. MEDIUM The hypotenuse is 15 and one angle is 37°. Find the side adjacent to the 37° angle.

$\cos 37° = \frac{x}{15},\ 0.8 = \frac{x}{15},\ x = 15 \times 0.8 = 12.$ (A 9-12-15 triangle — a scaled 3-4-5!)

17. MEDIUM A tree casts a shadow 12 m long. From the tip of the shadow, the top of the tree is at an angle of 53°. How tall is the tree?

The shadow is adjacent to the 53° angle; the tree is opposite. $\tan 53° = \frac{h}{12},\ h = 12 \times 1.33 \approx 16$ m.

18. HARD The side opposite a 30° angle is 7. Find the hypotenuse.

$\sin 30° = \frac{7}{h},\ 0.5 = \frac{7}{h},\ h = \frac{7}{0.5} = 14.$ Check: $\frac{7}{14} = 0.5$ ✓

19. HARD A kite string is 50 m long and makes a 37° angle with the ground. How high is the kite? How far away along the ground is it?

Height (opposite): $50 \times \sin 37° = 50 \times 0.6 = 30$ m. Ground distance (adjacent): $50 \times \cos 37° = 50 \times 0.8 = 40$ m. (30-40-50: a scaled 3-4-5 triangle.)

F. Think about it HARD

20. Why can $\sin A$ never be bigger than 1?

> $\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$ and the hypotenuse is the longest side, so the top is smaller than the bottom: the fraction is less than 1.

21. In a right triangle, $\sin A = \frac{3}{5}$ and the hypotenuse is 25. Find the opposite side. Then use $\cos A = \frac{4}{5}$ to find the adjacent side.

> Opposite $= \frac{3}{5} \times 25 = 15$. Adjacent $= \frac{4}{5} \times 25 = 20$. (15-20-25 is the 3-4-5 triangle scaled by 5.)

22. Why is $\tan 45^\circ$ exactly 1?

> The angles are $90^\circ, 45^\circ, 45^\circ$, so two angles are equal and the two legs are equal. $\tan 45^\circ = \frac{\text{opp}}{\text{adj}} = \frac{\text{same}}{\text{same}} = 1$.